

2024, Mar. 2024 Release LOC:Acute Pediatric Carbon Monoxide Poisoning

Overview

Select Day

Episode Day 1

Episode Day 2

Episode Day 3

Episode Day 4

Discharge Screens ⁽³³⁾

Notes

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Overview

Instruction: This subset is for patients with accidental or intentional carbon monoxide poisoning.

Introduction:

Carbon monoxide (CO) is an odorless, colorless, and nonirritating gas and carbon monoxide (CO) poisoning often presents with nonspecific signs and symptoms. It is estimated that 1,000-2,000 accidental deaths occur yearly from carbon monoxide inhalation. Poisoning occurs when the carbon monoxide binds with hemoglobin, impairing the delivery of oxygen to tissues. It also binds with myoglobin, leading to increased hypoxia in cardiac muscle and causing platelet and neutrophil activation, free radical formation, and lipid peroxidation in the brain and other tissues. Symptoms of carbon monoxide poisoning include: vague complaints of headache, dizziness, or nausea, to severe symptoms of hypotension, cardiac ischemia, respiratory failure, coma, and death. Neurologic sequelae include memory loss, difficulty with concentration or language, and long term impairments, such as, depression and parkinsonism. Carbon monoxide poisoning in a pregnant person can have severe adverse effects on the fetus. Carbon monoxide crosses the placental barrier and fetal hemoglobin has higher affinity for carbon monoxide. Morbidity and mortality from carbon monoxide poisoning is significant in a fetus. A mild maternal poisoning may present as severe poisoning in the fetus and death may occur.

Evaluation and treatment:

Carbon monoxide is considered toxic when the carboxyhemoglobin level is over 10% and severe poisoning occurs when levels are greater than 20% to 25%. People living in high pollution areas may have baseline levels of 3%-5% and heavy smokers may have a baseline of up to 15%.

Due to the risk of cardiac damage, all patients presenting with carbon monoxide poisoning should receive an electrocardiogram, cardiac biomarkers, and creatine phosphokinase (CPK). These patients should also be observed for signs of cerebral hemorrhage or edema.

The treatment for carbon monoxide poisoning is 100%(1.00) oxygen or hyperbaric oxygen therapy (HBOT), if available. Although HBOT is considered the standard of treatment among medical practitioners, the clinical trial results are inconclusive in the prevention of neurological sequelae after carbon monoxide poisoning.

Reference: (American College of Emergency Physicians Clinical Policies Subcommittee on Carbon Monoxide et al., Ann Emerg Med 2017; 69(1): 98-107 e106; Brown et al., Annals of Emergency Medicine 2017, 69: 98-107.e6)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed

for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽¹⁾

- Carboxyhemoglobin 25-29%(0.25-0.29) without mental status change ^(2, 3)
- Continuous cardiac monitoring (excludes Holter) ⁽⁴⁾
- Oxygen 100%(1.00) ^(5, 6)

● **INTERMEDIATE, All:** ⁽⁷⁾● **Finding, One:**

- Carboxyhemoglobin $\geq$ 30%(0.30) without mental status change ^(2, 3)
- Carboxyhemoglobin 25-29%(0.25-0.29) with mental status change ^(2, 3)
- Carboxyhemoglobin $>$ 15%(0.15) and pregnant ⁽⁸⁾

● **Intervention, $\geq$ One:**

- Hyperbaric oxygen therapy (HBOT) ⁽⁶⁾
- Oxygen 100%(1.00) ^(5, 6)
- Continuous cardiac monitoring (excludes Holter) ⁽⁴⁾

● **CRITICAL, $\geq$ One:** ⁽⁹⁾● **Carbon monoxide poisoning, Both:**● **Finding, One:**

- Carboxyhemoglobin $\geq$ 30%(0.30) with mental status change ^(2, 3)
- Carboxyhemoglobin $\geq$ 40%(0.40) ^(3, 10)

● **Intervention, $\geq$ One:**

- Hyperbaric oxygen therapy (HBOT) ⁽⁶⁾
- Oxygen 100%(1.00) ^(5, 6)
- Mechanical ventilation
- NIPPV ⁽¹¹⁾

● **Episode Day 2, One:**● **OBSERVATION, One:**● **Responder, discharge expected today if clinically stable last 12h, All:** ⁽¹²⁾

- O₂ sat $\geq$ 94%(0.94) or within acceptable limits ⁽¹³⁾
- Carboxyhemoglobin $<$ 10%(0.10) or within acceptable limits ^(3, 10, 13)
- Neurological stability ⁽¹⁴⁾

● **Non-responder, not clinically stable for discharge and exceeds initial 24h Observation for this condition, One:**

- For carbon monoxide poisoning use Episode Day 2 at a higher level of care
- For a condition other than carbon monoxide poisoning use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

● **ACUTE, One:** ⁽¹⁵⁾● **Responder, discharge expected today if clinically stable last 24h, All:** ⁽¹²⁾

- O₂ sat $\geq$ 94%(0.94) or within acceptable limits ⁽¹³⁾
- Carboxyhemoglobin $<$ 10%(0.10) or within acceptable limits ^(3, 10, 13)
- Neurological stability ⁽¹⁴⁾

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽¹⁶⁾

● **Complication or comorbidity and clinically stable for discharge, One:**● **Functional impairment and, One:** ⁽¹⁷⁾● **Discharge to home and, One:** ⁽¹⁸⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services

● **Partial responder, not clinically stable for discharge and requires continued stay, $\geq$ One:**

- Neurological impairment or condition and, **Both:** ⁽¹⁹⁾
 - Finding, ≥ **One:**
 - Acute onset
 - Deterioration
 - Neurological assessment at least $3x/24h \leq 2d$ ⁽²⁰⁾
- Functional impairment (new) requiring rehabilitation initiation $\leq 24h$ and, ≥ **One:** ^(17, 21)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
 - Post Critical care $\leq 24h$
- **INTERMEDIATE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽¹²⁾
 - O_2 sat $\geq 92\%(0.92)$ or within acceptable limits ⁽¹³⁾
 - Carboxyhemoglobin $< 10\%(0.10)$ or within acceptable limits ^(10, 13)
 - Neurological stability ⁽¹⁴⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Acute cerebral hemorrhage, thrombosis, or infarct, actual or suspected (**use General Medical subset**)
 - Troponin positive or ACS suspected (**use General Medical subset**) ⁽²²⁾
 - Arrhythmia, ≥ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk $\leq 24h$
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) $\leq 3d$ ⁽²³⁾
 - Neurological impairment or condition and, ≥ **One:** ⁽¹⁹⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h $\leq 2d$ ⁽²⁰⁾
 - NIPPV ⁽¹¹⁾
 - Oxygen 35-39%(0.35-0.39)
- **CRITICAL, ≥ One:**
 - Acute cerebral hemorrhage, thrombosis, or infarct, actual or suspected (**use General Medical subset**)
 - Troponin positive or ACS suspected (**use General Medical subset**) ⁽²²⁾
 - Arrhythmia, ≥ **One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - Antiarrhythmic and, **Both:**
 - Arrhythmia, ≥ **One:**
 - Recurrent arrhythmia ⁽²⁴⁾
 - Refractory arrhythmia ⁽²⁴⁾
 - Symptomatic, ≥ **One:**
 - Chest pain
 - Heart failure and, ≥ **One:**
 - Arterial $PO_2 < 56$ mmHg(7.4 kPa)
 - O_2 sat $< 89\%(0.89)$ and $<$ baseline ⁽²⁵⁾
 - Hemodynamic instability, ≥ **One:**
 - Heart rate, **One:** ⁽²⁶⁾
 - $> 180/min$, sustained (age ≤ 12 mos) ⁽²⁷⁾
 - $> 160/min$, sustained (age > 12 mos to 4 yrs) ⁽²⁷⁾
 - $> 150/min$, sustained (age > 4 yrs to 8 yrs) ⁽²⁷⁾
 - $> 140/min$, sustained (age > 8 yrs to 15 yrs) ⁽²⁷⁾
 - $> 130/min$, sustained (age > 15 yrs to < 18 yrs) ⁽²⁷⁾

- **Systolic BP, One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁸⁾
- **MAP, One:** ⁽²⁹⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
- **Cerebral edema and, ≥ One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid
- **Hyperosmolar therapy, ≥ One:** ⁽³⁰⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Mechanical ventilation
- **Neurological impairment or condition, Both:** ⁽¹⁹⁾
 - Acute onset or deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽²⁰⁾
- **NIPPV and, ≥ One:** ⁽¹¹⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽³²⁾
- Oxygen ≥ 40%(0.40)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- (this criteria replaces Acute hypertension criteria)**
- **Episode Day 3, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽¹²⁾
 - O₂ sat ≥ 94%(0.94) or within acceptable limits ⁽¹³⁾
 - Carboxyhemoglobin < 10%(0.10) or within acceptable limits ^(3, 10, 13)
 - Neurological stability ⁽¹⁴⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹⁶⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - **Functional impairment and, One:** ⁽¹⁷⁾
 - **Discharge to home and, One:** ⁽¹⁸⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Neurological impairment or condition and, Both:** ⁽¹⁹⁾
 - **Finding, ≥ One:**
 - Acute onset
 - Deterioration
 - Neurological assessment at least 3x/24h ≤ 2d ⁽²⁰⁾
 - **Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ One:** ^(17, 21)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
 - Post Critical care ≤ 24h
 - **INTERMEDIATE, One:**

- **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽¹²⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽¹³⁾
 - Carboxyhemoglobin < 10%(0.10) or within acceptable limits ^(10, 13)
 - Neurological stability ⁽¹⁴⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Acute cerebral hemorrhage, thrombosis, or infarct, actual or suspected (**use General Medical subset**)
 - Troponin positive or ACS suspected (**use General Medical subset**) ⁽²²⁾
- **Arrhythmia, ≥ One:**
 - IV antiarrhythmic and hemodynamically stable
 - **Antiarrhythmic conversion from IV to PO, One:**
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽²³⁾
- **Neurological impairment or condition and, Both:** ⁽¹⁹⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h ≤ 2d ⁽²⁰⁾
- NIPPV ⁽¹¹⁾
- Oxygen 35-39%(0.35-0.39)
- **CRITICAL, ≥ One:**
 - Acute cerebral hemorrhage, thrombosis, or infarct, actual or suspected (**use General Medical subset**)
 - Troponin positive or ACS suspected (**use General Medical subset**) ⁽²²⁾
- **Arrhythmia, ≥ One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
- **Antiarrhythmic and, Both:**
 - **Arrhythmia, ≥ One:**
 - Recurrent arrhythmia ⁽²⁴⁾
 - Refractory arrhythmia ⁽²⁴⁾
 - **Symptomatic, ≥ One:**
 - Chest pain
 - **Heart failure and, ≥ One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa)
 - O₂ sat < 89%(0.89) and < baseline ⁽²⁵⁾
 - **Hemodynamic instability, ≥ One:**
 - **Heart rate, One:** ⁽²⁶⁾
 - > 180/min, sustained (age ≤ 12 mos) ⁽²⁷⁾
 - > 160/min, sustained (age > 12 mos to 4 yrs) ⁽²⁷⁾
 - > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽²⁷⁾
 - > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽²⁷⁾
 - > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽²⁷⁾
 - **Systolic BP, One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁸⁾
 - **MAP, One:** ⁽²⁹⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
- **Cerebral edema and, ≥ One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid
- **Hyperosmolar therapy, ≥ One:** ⁽³⁰⁾
 - Mannitol

- Hypertonic saline $\geq 3\%$ (0.03) solution
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Mechanical ventilation
- Neurological impairment or condition, **Both:** ⁽¹⁹⁾
 - Acute onset or deterioration
 - Neurological assessment every 1-2h, $\leq 2d$ ⁽²⁰⁾
- NIPPV and, $\geq$ **One:** ⁽¹¹⁾
 - $FiO_2 \geq 40\%$ (0.40)
 - Respiratory intervention every 1-2h ⁽³²⁾
- Oxygen $\geq 40\%$ (0.40)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- (this criteria replaces Acute hypertension criteria)**
- **Episode Day 4, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽¹²⁾
 - O_2 sat $\geq 92\%$ (0.92) or within acceptable limits ⁽¹³⁾
 - Carboxyhemoglobin $< 10\%$ (0.10) or within acceptable limits ^(10, 13)
 - Neurological stability ⁽¹⁴⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹⁶⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - **Functional impairment and, One:** ⁽¹⁷⁾
 - **Discharge to home and, One:** ⁽¹⁸⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, $\geq$ **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Responder**, discharge expected today if clinically stable last 24 hours, **All:** ⁽¹²⁾
 - O_2 sat $\geq 94\%$ (0.94) or within acceptable limits ⁽¹³⁾
 - Carboxyhemoglobin $< 10\%$ (0.10) or within acceptable limits ^(10, 13)
 - Neurological stability ⁽¹⁴⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽³³⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽³⁴⁾
 - Patient or caregiver demonstrates ability to manage care
 - **Carbon monoxide (CO) poisoning discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽³⁵⁾
 - Education on the importance of preventing carbon monoxide (CO) poisoning ⁽³⁶⁾

- Medication reconciliation ⁽³⁷⁾
- Follow-up care planned ⁽³⁸⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● **Level of care appropriateness, All:**

- Home environment safe and accessible ⁽³⁴⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(39, 40)

● **Skilled services, ≥ One:** ⁽⁴¹⁾

- Skilled nursing ⁽⁴²⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴³⁾

● **Carbon monoxide (CO) poisoning discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁵⁾
- Education on the importance of preventing carbon monoxide poisoning ⁽³⁶⁾
- Medication reconciliation ⁽³⁷⁾
- Follow-up care planned ⁽³⁸⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

● **Level of care appropriateness, One:**

- Support systems available ⁽⁴⁴⁾

● **Skilled treatment, ≥ One:**

- Clinical assessment ⁽⁴⁵⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation

● **SKILLED MEDICAL OR THERAPY, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● **Services, ≥ One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴²⁾

● **Therapy, All:**

- Goal directed therapy with rehabilitation potential ^(46, 47)
- Functional impairments requiring at least supervision ⁽⁴⁸⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽³⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
- Treatment precluded in a lower level

● **Services, ≥ One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴²⁾

● **Therapy, All:**

- Goal directed therapy with rehabilitation potential ^(46, 47)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁸⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽³⁷⁾
- Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE REHAB, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ One:

- Goal directed therapy with rehabilitation potential ^(46, 47)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁸⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽³⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● ACUTE REHAB, Both:**● Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, All:

- Goal directed therapy with rehabilitation potential ^(46, 47)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁸⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁴⁹⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽³⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● LONG TERM ACUTE CARE, Both:**● Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵⁰⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(51, 52)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵³⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽³⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

For patients at the Acute level of care, mental status change may include disorientation, agitation, increasing irritability, listlessness, or lethargy. The inability to soothe an infant, abnormal cry (e.g., weak, high-pitched), or decreased social interaction with parents or caregivers in children 2 years old or younger, may also qualify as a mental status change (Keith Kleinman, In: Harriet Lane Handbook, 22nd edn. 2021).

Instruction: Patients with acute coma, stupor, or somnolence should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma or stupor.

3:

Although the diagnosis of carbon monoxide (CO) poisoning can be confirmed by detecting increased levels of carboxyhemoglobin in the blood, the presence of clinical signs and symptoms after a known exposure should not be ignored. Carboxyhemoglobin levels over 10% are considered to be toxic. Severe poisoning occurs when levels exceed 20% with symptoms of cardiac or cerebral ischemia (Guzman, Critical Care Clinics 2012, 28: 537-48).

4:

Carbon monoxide (CO) poisoning frequently causes nonspecific repolarization and arrhythmias, often resulting in abnormal electrocardiography findings. Cardiotoxicity may not be apparent in the presence of exacerbated neurological findings (i.e., cerebral hemorrhage or edema) or respiratory symptoms. There is a high risk of cardiac damage associated with carbon monoxide (CO) poisoning and all patients presenting with an actual or suspected poisoning should receive an electrocardiogram (ECG) and cardiac biomarkers (Brown et al., Annals of Emergency Medicine 2017, 69: 98-107.e6).

5:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO₂] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO₂) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO₂ optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FIO₂

Reservoir systems (For neonates, infants, and young children, the maximum FIO₂ achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FIO₂
- Partial-rebreathing masks 40-60%(0.40-0.60) FIO₂
- Non-rebreathing masks > 60%(0.60) FIO₂

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FIO₂
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FIO₂

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FIO₂

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

6:

In all cases of carbon monoxide (CO) poisoning, high-flow oxygen by mask or endotracheal tube is the first line of treatment. If available, hyperbaric oxygen therapy (HBOT) can be used in the treatment of carbon monoxide poisoning. During HBOT, the oxygen displaces the carbon monoxide from the hemoglobin. There is insufficient evidence to demonstrate the superior efficacy with HBOT when compared to normobaric 100% oxygen, however, it continues to be the standard of treatment (American College of Emergency Physicians Clinical Policies Subcommittee on Carbon Monoxide et al., Ann Emerg Med 2017; 69(1): 98-107 e106; Brown et al., Annals of Emergency Medicine 2017, 69: 98-107.e6).

7:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

8:

Carbon monoxide (CO) poisoning in the pregnant individual can have severe adverse effects on the fetus. Fetal hemoglobin has a higher affinity for carbon monoxide and crosses the placental barrier. Morbidity and mortality from carbon monoxide poisoning is significant in a fetus. A mild maternal poisoning may present as severe poisoning in the fetus and death may occur (Smollin and Olson, Clin Evid, 2010).

9:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

10:

Patients that live in high pollution areas may have a baseline carboxyhemoglobin of 3-5%(0.03-0.05), while heavy smokers may have a baseline carboxyhemoglobin of up to 15%(0.15) (Brown et al., Annals of Emergency Medicine 2017, 69: 98-107.e6).

11:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin

North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

12:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

13:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

14:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

15:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

16:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

17:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

18:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

19:

Neurological deficit(s) may include:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

Neurological disorder may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

20:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

21:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

22:

Cardiac biomarkers are used to detect myocardial injury. These markers assist in the diagnosis or exclusion of a myocardial infarction and must be interpreted in conjunction with the patient's symptoms, examination, and electrocardiogram findings. Cardiac biomarkers are assessed in all patients with suspected acute coronary syndrome and should include a cardiac troponin assay. Compared to other cardiac biomarkers, troponin assays have high specificity and sensitivity when used serially to identify myocardial cell death. Troponin assays can detect the majority of myocardial infarctions within 3 hours of presentation. In some cases, detection may be delayed up to 12 hours after the onset of symptoms and repeat levels are recommended at 8 to 12 hours after symptom onset if initial tests are negative. Cardiac biomarker levels defining a positive value are determined by laboratory protocol and may vary between hospitals (Thygesen et al., Journal of the American College of Cardiology 2018, 72: 2231-64; Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

Instruction: While there are a variety of medical conditions that can lead to troponin elevation, the intent of this criterion is to capture patients with positive troponins due to acute coronary syndrome. Those patients with elevated troponins stemming from demand ischemia should be reviewed in the subset that corresponds to the source of their demand (e.g., pneumonia, heart failure, renal failure, gastrointestinal bleeding, etc.).

23:

Various antiarrhythmics with proarrhythmic risk are administered to the pediatric population. Initiation or dose adjustments of these medications require monitoring in the inpatient setting. Some but not all of these medications include (Kusumoto et al., J Am Coll Cardiol 2018, 72: 1653-76):

- Sotalol
- Amiodarone
- Disopyramide
- Flecainide

Many factors influence the appropriate choice of drug including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

24:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

25:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

26:

In the pediatric patient, hemodynamic instability could be assessed by an elevated heart rate (HR) above the 95th percentile norms based on specific age groups. A combination of different parameters and their trends could also

be utilized to ascertain impending clinical deterioration more accurately in these patients to allow for closer monitoring in the critical care setting, including respiratory rate, mental status, and capillary refill time, amongst others. Other factors that could impact HR in the pediatric population include pain, anxiety, medications, and/or fever. Although no specific consensus has been reached on HR thresholds that signify unstable hemodynamics by various pediatric societies, systemic reviews and observational studies have compared HR data to textbook references such as the Pediatric Advance Life Support (PALS). Eytan et al., in 2017, studied HR in pediatric patients who were critically ill in the pediatric intensive care unit (PICU) and developed percentiles of heart rates categorized into the same age groups used in previous studies, including PALS, with the goal of comparing their observed norms, especially between the 5th and 95th percentiles (Eytan et al., *Frontiers in Pediatrics* 2017, 5). Using their HR data above the 95th percentile, HR thresholds were utilized to identify the pediatric patient who may require immediate or urgent intervention and rapidly treat unstable hemodynamics to prevent a potentially serious event.

27:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

28:

Blood pressure is considered labile when it is extremely variable and difficult to control.

29:

In pediatric patients, hemodynamic instability may be defined as a mean arterial pressure (MAP) less than 55 mmHg in infants up to 1 year of age and less than 65 mmHg in patients over 1 year of age (Matics and Sanchez-Pinto, *JAMA Pediatr* 2017, 171: e172352).

30:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., *Neurocrit Care* 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

31:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., *Cochrane Database Syst Rev* 2013, 2: CD003408).

32:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

33:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

34:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

35:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

36:

All homes should have carbon monoxide detectors installed and tested on a regular basis. Measures to prevent carbon monoxide poisoning in the home include:

- Make sure household appliances are safe and well maintained
- Never use ovens or gas ranges to heat your home
- Rooms should be well ventilated
- Chimneys and flues should be cleaned at least once a year
- Do not use gas powered equipment in your home
- Do not burn charcoal indoors

37:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

38:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

39:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

40:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

41:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

42:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

43:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house

extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

44:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

45:

The initial clinical assessment is performed by a licensed professional, which includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems are identified. The clinical assessment includes:

- Patient history, physical exam, vital signs, height and weight
- Pain assessment includes: cause, intensity, quality, onset, duration, effects on quality of life with activities of daily living (ADLs), instrumental activities of daily living (IADLs), play, and interpersonal relationships, including a current pain management regimen.
- Patient's nutritional and hydration status
- Effects of current illness or injury on age appropriate function
- Safety and infection control measures
- Prescription and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient and family's coping strategies to deal with the illness or injury and ability to follow the plan of care

46:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

47:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage,

- percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

48:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

49:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

50:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

51:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

52:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure

- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

53:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.